COMP 7010 Advanced Algorithms — Assignment 3: Linear Programming

Fall 2026 · Prof. Salem Othman · Out Tuesday Nov 3 · Due Tuesday Nov 17, 6:00 PM (start of class)

Overview

This assignment covers Weeks 7–8: writing problems as linear programs, the simplex method, and duality. It has three parts worth 100 points. Submit **two separate files** in Brightspace: `LastName_A3.pdf` (the report) and `LastName_A3.zip` (all code plus a README). Do not put the PDF inside the ZIP.

Part	What	Points
A	Implementation: your own simplex method, with dual prices and a certificate check	40
B	Modeling and measurement: a roasting plan, and how many pivots simplex needs	30
C	By hand: simplex pivots, the dual, standard form	30

Language: any. Python is assumed below; if you use another language, keep the same function names and behavior. You may use `scipy.optimize.linprog` **only to check** your answers, never inside your own solver.

Part A — Implementation (40 points)

A1. The simplex method (25 points)

Write `simplex(c, A, b, rule="bland")` for the problem

maximize $c^T x$ subject to $Ax \le b$, $x \ge 0$, with every $b_i \ge 0$

(so the plan x = 0 is an allowed starting corner). It must return a tuple `(status, x, z, y, pivots)`:

- `status` — `"optimal"` or `"unbounded"`
- `x` — the best plan (list of numbers), `z` — the best value
- `y` — the dual prices, one per constraint, read from the final profit row under the slack columns
- `pivots` — the number of pivots performed

Requirements:

1. Use the tableau method from class: start with one slack per row; most-negative (or chosen) entry of the profit row enters; ratio test over rows with a **positive** entry only; divide the pivot row, then clear the column.
2. Support two entering rules: `rule="dantzig"` (most negative entry of the profit row) and `rule="bland"` (smallest index among the negative entries; break ratio-test ties by the smallest index of the leaving variable). (5 of the 25 points)
3. Use exact arithmetic (`fractions.Fraction`), so 1/3 stays 1/3.
4. Raise an error if some $b_i < 0$ (that would need Phase I, which you do not have to write).
5. Do not use any library that solves LPs for you.

Run it on these five tests and include the output in your report:

Test	c	A	b	Expected
Bakery (class)	[30, 20]	[[1,1],[1,3],[1,0]]	[4, 9, 3]	z = 110, x = (3, 1), y = (20, 0, 10)
Print shop (class)	[5, 4]	[[6,4],[1,2]]	[24, 6]	z = 21, x = (3, 3/2), y = (3/4, 1/2)
Example B (class)	[3, 5]	[[1,0],[0,2],[3,2]]	[4, 12, 18]	z = 36, x = (2, 6)
Degenerate	[30, 20]	[[1,1],[1,3],[1,0]]	[4, 6, 3]	z = 110 (a ratio-test tie)
Unbounded	[1, 1]	[[1,−1]]	[1]	status “unbounded”

A2. Certificate check (5 points)

Write `check_certificate(c, A, b, x, y)` that returns `True` only if **all three** hold, using exact arithmetic:

1. x is an allowed plan ($Ax \le b$, $x \ge 0$);
2. y is an allowed price list for the dual ($A^T y \ge c$, $y \ge 0$);
3. $c^T x = b^T y$.

By strong duality, these three facts together prove x is optimal without trusting your solver. Call it on every optimal result in A1 and A3.

A3. Random test against SciPy (10 points)

Generate 20 random LPs with n and m between 2 and 6, entries of A from 1 to 9, b from 10 to 50, c from 1 to 20. For each one, solve it with your `simplex` and with `linprog` (remember: `linprog` minimizes), and assert that

- the two best values agree to within 10^{-9}, and
- `check_certificate` returns `True`.

Report how many of the 20 passed (it should be 20).

Part B — Modeling and measurement (30 points)

B1. A coffee roaster (12 points)

A small roaster makes three blends per week, in batches. Fractions of a batch are allowed.

	House	Dark	Decaf	Available per week
Profit per batch	$5	$8	$6	—
Roaster time (hours)	3	1	2	24
Green beans (kg)	4	3	4	48
Packing time (hours)	1	4	2	20

1. Write the LP: variables with units, objective, constraints. (3 points)
2. Solve it with your `simplex`. Report the plan, the profit and the number of pivots. (2 points)
3. Report the dual prices. For each resource, say in one sentence what its price means and the most the roaster should pay for one extra unit of it (one more roaster hour, kg of beans, or packing hour). (3 points)
4. Dark has the highest profit per batch. Use the dual prices to explain in two sentences why the best plan makes **no** Dark. (2 points)
5. Write the dual LP and verify with `check_certificate` that the two optimal values are equal. (2 points)

B2. How many pivots? The Klee–Minty problem (12 points)

For n variables, the Klee–Minty LP is

maximize the sum over j = 1..n of $2^{(n-j)} \cdot x_j$

subject to, for each row i = 1..n: $2 \cdot (\text{sum over } j < i \text{ of } 2^{(i-j)} \cdot x_j) + x_i \le 5^i$, and $x \ge 0$.

(For n = 3: maximize $4x_1 + 2x_2 + x_3$ subject to $x_1 \le 5$; $4x_1 + x_2 \le 25$; $8x_1 + 4x_2 + x_3 \le 125$.)

1. Write a function that builds c, A, b for any n. Check your n = 3 against the example above. (2 points)
2. For n = 2, 3, …, 8, run your simplex with `rule="dantzig"` and with `rule="bland"`. Make a table of the pivot counts and the optimal values. (4 points)
3. Plot pivots against n for both rules, with a logarithmic y-axis, labeled axes and a legend. (3 points)
4. In three sentences: what formula fits the Dantzig counts, what this says about the worst case of simplex, and why this does not contradict "LP can be solved in polynomial time". (3 points)

B3. Random problems (6 points)

For m = n = 5, 10, 20, 30, 40, generate 5 random LPs each (A entries 1–9, b entries from 10m to 50m, c from 1 to 20). Run both rules and report the average number of pivots per size in a table and one plot. In two sentences, compare with B2.

Part C — By hand (30 points)

Solve these without software: write each step yourself, either handwritten (photographed neatly) or typed, and include it in the PDF. Use fractions, not decimals.

C1. Simplex by hand (12 points)

maximize $4x + 3y$ subject to $x + y \le 5$, $2x + y \le 8$, $x + 3y \le 12$, $x, y \ge 0$.

1. Write the starting dictionary (or tableau) with slacks s1, s2, s3. (2 points)
2. Do every pivot. For each one write: which variable enters and why, the ratio test with every limit, which variable leaves, and the new dictionary. (6 points)
3. Explain why you stop, using the final profit row. (2 points)
4. Draw the feasible region, mark the corners, and mark the path simplex took. (2 points)

C2. The dual of C1 (10 points)

1. Write the dual LP of C1, naming the dual variables p, q, r. (3 points)
2. Read the dual prices from your final dictionary in C1. (2 points)
3. Find the same prices again using complementary slackness only: say which price must be 0 and why, and which dual constraints must be exact and why, then solve. (3 points)
4. Show that the dual value equals the primal value. (2 points)

C3. Standard form (8 points)

minimize $x_1 + 2x_2$ subject to $x_1 + x_2 \geq 3$, $x_1 - x_2 = 1$, $x_1 \geq 0$, x_2 free (may be negative).

1. Convert it to standard form (maximize, all ≤, all variables ≥ 0). Show one fix per step and say which fix you are using. (4 points)
2. Solve the original by hand (hint: use the equality to remove x_1) and give the optimal plan and value. (2 points)
3. Explain in two sentences why the simplex method cannot start at the all-zero plan of your standard form, and what it needs to do instead. (2 points)

Report format

Your PDF should contain, in order: Part A test output and a short note on your design; Part B tables, plots and answers; Part C hand work. Keep it under 8 pages. Put your code and a README that says how to run each part in the ZIP; submit the PDF as its own file.

Policies (from the syllabus)

- AI tools may be used to explain concepts, explore approaches, debug error messages, and review your own code. They may not be used to generate the solutions. Any AI-assisted portion must be labeled in a comment or footnote stating the tool and how it was used.
- Late work: 10% penalty per day. Extensions must be requested before the due date.
- This is individual work. You may discuss ideas with classmates; the code, the plots and the hand work must be your own.

Grading rubric

Component	Points	Full credit means
A1 simplex	25	Correct on all five tests (status, x, z, y, pivots printed); both rules; exact fractions; ratio test uses positive entries only
A2 certificate	5	Checks all three conditions exactly; called on every optimal result
A3 random test	10	20 random LPs, all agree with SciPy and pass the certificate
B1 roaster	12	Correct LP, plan, profit and prices; correct meaning of each price; clear reason for no Dark; dual written and checked
B2 Klee–Minty	12	Correct construction; full table for n = 2..8; log-scale plot; correct formula and explanation
B3 random	6	Table and plot for all sizes; sensible comparison with B2
C1 simplex by hand	12	Every pivot correct with the ratio test shown; stopping reason; region and path drawn
C2 dual	10	Correct dual; prices read correctly; complementary slackness argument complete; equal values
C3 standard form	8	Each fix named and correct; correct optimum; Phase I reason